

Unsupervised Learning for Discovery

Clustering • PCA • DBSCAN • Interpretation • Validation • Limitations

Applied AI Design Workshop for Faculty
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Goal

discover structure in unlabeled data while avoiding over-interpretation.

Session outputs

- Unsupervised problem statement
- Scaled feature table
- Cluster interpretation plan
- Validation approach
- Limitation statement

Core idea

Clusters are mathematical groupings. They become meaningful only after domain interpretation and validation.

Session 4 learning outcomes

Distinguish

Explain supervised vs. unsupervised learning and when labels are absent

Cluster

Describe k-means, hierarchical clustering, and DBSCAN at a conceptual level

Prepare

Scale variables and choose features before clustering

Validate

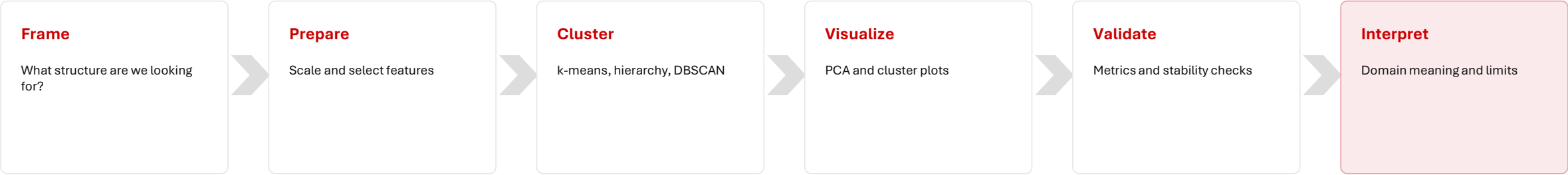
Use metrics, visual inspection, and domain checks to assess structure

Interpret

Translate clusters into hypotheses without claiming causality or truth

Participant output: a clustering plan that includes features, scaling, validation, interpretation, and limitations.

Activity roadmap



Facilitator note

The point is not to “find the right clusters.” The point is to generate defensible hypotheses about structure in unlabeled data.

Unsupervised learning supports discovery, exploration, segmentation, and data-quality investigation.

What makes a problem unsupervised?

Unsupervised learning searches for structure without a known target label.

Supervised learning

Known target label:

- strength value
- defect / no defect
- pass / fail
- diagnosis

Question: can inputs predict the target?

Unsupervised learning

No target label:

- unknown groups
- operating regimes
- abnormal cases
- latent patterns

Question: what structure exists?

Common use

Explore data before labels exist, identify natural groupings, or create hypotheses for follow-up experiments.

Unsupervised results are usually starting points for investigation, not final decisions.

Applied examples of unsupervised discovery

Domain	Data	Discovery question
Manufacturing	temperature, speed, vibration, energy	Are there distinct operating regimes?
Bioprocessing	pH, dissolved oxygen, agitation, OD trajectory features	Which batches behave similarly?
Education	assignment timing, submissions, interaction frequency	Are there patterns of student engagement?
Materials	composition, process settings, measured properties	Do material families emerge from the data?
Maintenance	sensor summaries from equipment cycles	Which cycles look atypical or degraded?

A good unsupervised question connects data patterns to a plausible disciplinary interpretation.

Part 1: prepare the feature table

Clustering is sensitive to feature choice and scale.

Choose features

Use variables that represent the phenomenon of interest. Remove identifiers and target-like columns.

Scale numeric data

Variables with large units can dominate distance-based methods. Standardize before k-means, PCA, and many distance methods.

Handle missing data

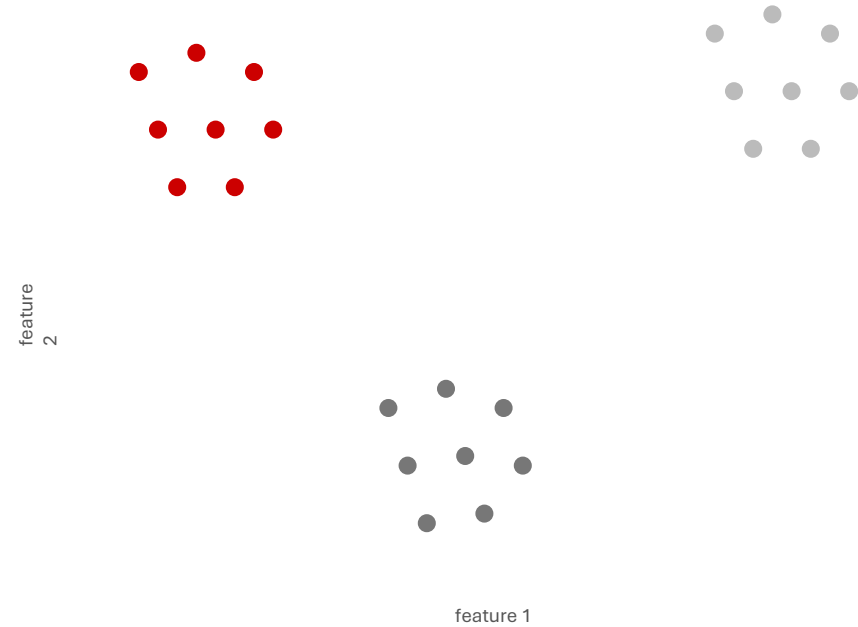
Impute carefully or remove variables with too much missingness. Missingness may itself carry meaning.

Guiding question

Do the selected features describe similarity in the way a domain expert would understand it?

k-means clustering: compact groups around centers

k-means partitions observations into K groups by minimizing distance to cluster centers.



Requires K

You must choose the number of clusters before fitting.

Works best when

Clusters are compact, separated, and roughly similar in size.

Sensitive to scale

Standardization strongly affects distance calculations.

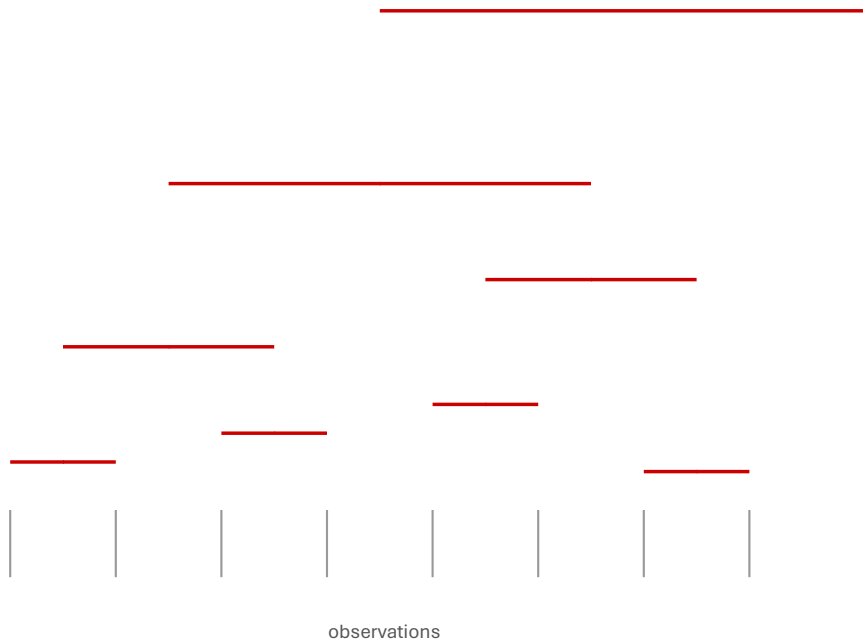
Interpretation risk

A cluster number does not automatically equal a real category.

Use k-means when a compact-center view of similarity is defensible.

Hierarchical clustering: nested similarity structure

Hierarchical methods show how observations merge into groups at different levels of similarity.



Strength

Reveals nested structure and does not require choosing K before fitting.

Choice point

Cut the tree at a level that balances detail and interpretability.

Key inputs

Distance metric and linkage method influence results.

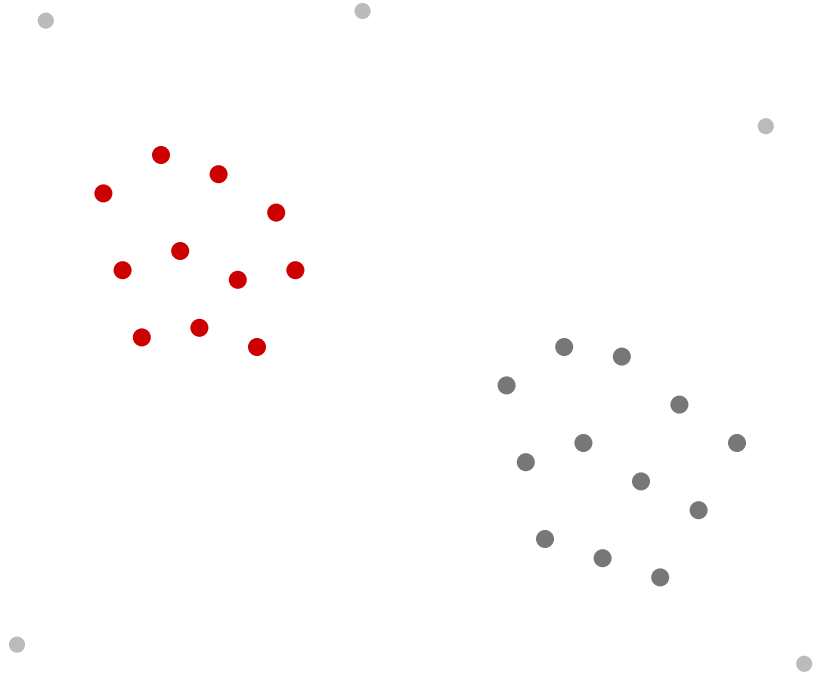
Caution

Can be hard to read for very large datasets without sampling or summarization.

Hierarchical clustering is useful when you want to see structure at multiple resolutions.

DBSCAN: dense groups and noise points

DBSCAN groups observations in dense regions and can label sparse points as noise.



Strength

Finds irregularly shaped clusters and identifies noise or outliers.

Parameters

eps controls neighborhood size; min_samples controls density requirement.

Works best when

Clusters differ from noise by density and the scale of distance is meaningful.

Caution

Can struggle when clusters have very different densities.

DBSCAN is useful when “unusual” observations matter as much as group membership.

PCA for visualization and simplification

Principal component analysis projects high-dimensional data into fewer dimensions.

What PCA does

Finds directions of maximum variation and creates new component axes.

Why use it here?

Visualize clusters, detect redundancy, and summarize many correlated features.

Important caution

PCA components are combinations of variables, not original physical quantities.

Component	Interpretation question
PC1	Which variables load strongly and define the main direction of variation?
PC2	What additional variation is captured beyond PC1?
Explained variance	How much of the original variation is retained?

PCA is a lens for exploration, not a guarantee of meaningful clusters.

How do we validate unsupervised results?

There is no single accuracy score because there is no target label.

Internal metrics

Silhouette score, within-cluster distance, cluster separation. Useful but not enough.

Stability checks

Do clusters persist under new samples, resampling, or small feature changes?

Domain validation

Do clusters align with known mechanisms, regimes, or follow-up measurements?

Key principle

A useful cluster should lead to a testable explanation or practical action.

Validation combines math, visualization, and domain evidence.

Interpreting clusters: from groups to hypotheses

Cluster labels such as 0, 1, and 2 are not explanations by themselves.

Step	Question to ask
Describe each cluster	What are the average feature values and ranges?
Compare to metadata	Do clusters differ by machine, batch, cohort, material, or time period?
Inspect examples	What do representative and borderline cases look like?
Name carefully	Use descriptive names only after evidence supports them.
Plan follow-up	What experiment, label review, or expert check would validate the hypothesis?

A cluster is a hypothesis generator; domain experts decide whether the pattern is meaningful.

Session 4 technical skill: minimal clustering workflow

Goal: scale features, run k-means, visualize clusters, and compute an internal metric.

```
import pandas as pd
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.decomposition import PCA
from sklearn.metrics import silhouette_score

features = ["temperature", "pressure", "vibration", "energy"]
X = df[features]

X_scaled = StandardScaler().fit_transform(X)

model = KMeans(n_clusters=3, random_state=42, n_init=10)
clusters = model.fit_predict(X_scaled)

df["cluster"] = clusters
print("Silhouette:", silhouette_score(X_scaled, clusters))

pca = PCA(n_components=2)
coords = pca.fit_transform(X_scaled)
```

Notebook

Use the Session 4 unsupervised learning notebook or embedded technical notebook.

Instructor prompt

“What evidence would convince you these clusters are meaningful?”

Coding focus

Scale → cluster → visualize → interpret → validate.

Faculty activity: design an unsupervised analysis

Use an unlabeled dataset or remove the target from a known dataset.

Decision area	Participant response
Discovery question	
Unit of observation	
Features used for similarity	
Features removed and why	
Scaling or preprocessing plan	
Candidate method: k-means, hierarchy, DBSCAN	
Validation approach	
What would make a cluster meaningful?	

Deliverable: one unsupervised learning plan with interpretation and validation criteria.

Classroom discussion prompts

- What does it mean for two observations to be similar?
- Which variables should define similarity — and which should not?
- How many clusters would be useful rather than merely mathematical?
- Do the clusters persist if features or scaling change?
- Are clusters aligned with meaningful metadata or known regimes?
- What follow-up evidence would validate the interpretation?
- What should not be concluded from the clusters?

Synthesis

Unsupervised learning helps faculty explore structure, generate hypotheses, and guide data collection — but it does not replace labels, experiments, or domain judgment.

The most important output is not a cluster label — it is a defensible interpretation plan.

Session 4 wrap-up: what you should take forward

Before clustering

- Define the discovery question
- Choose the unit of observation
- Select meaningful features
- Remove IDs and leakage variables
- Scale numeric data

During clustering

- Compare simple methods
- Visualize with PCA
- Check sensitivity to K and features
- Inspect representative cases
- Record assumptions

After clustering

- Describe cluster profiles
- Validate with metadata or experts
- Avoid causal claims
- Plan follow-up labeling
- State limitations clearly

Day 4 endpoint: faculty can complete a baseline supervised workflow and design an unsupervised discovery analysis.